Direct vs. Fragmentation Photon Composition at Truth in Pythia: Isolation-Cut Dependence

PPG12 Analysis (automated investigation)

April 18, 2026

Abstract

At truth level in the PPG12 signal Monte Carlo (Pythia 8, `photon5`+`photon10`+`photon20`), the direct photon fraction $f_{\text{direct}} = N_{\text{direct}}/(N_{\text{direct}} + N_{\text{frag}})$ is extracted as a function of the truth isolation cut at $R = 0.3$. The classification uses the HepMC-traced label `particle_photonclass` already present in the slimtree (class 1 = direct 2→2 hard scatter, class 2 = fragmentation 1→2 parton splitting; filled in `CaloAna24.cc`). At the PPG12 fiducial cut $E_T^{\text{iso,truth}} < 4$ GeV, $f_{\text{direct}} = 0.746,\ 0.793,\ 0.818$ in truth-photon p_T bins $[8, 14)$, $[14, 22)$, $[22, 36)$ GeV; the complement, $f_{\text{frag}} \approx 0.25{-}0.18$, is *not* negligible. The iso-cut dependence is weak: relaxing the cut from 4 GeV to no cut at all shifts f_{direct} by only ~ 0.8 percentage points, and tightening to < 1 GeV shifts it by ~ 1.5. The nominal efficiency denominator (`RecoEffCalculator.C`, photonclass<3) correctly includes both direct and fragmentation, so the reported cross section is strictly the isolated prompt (= direct + fragmentation) photon cross section at $E_T^{\text{iso,truth}} < 4$ GeV — as intended. No change to the nominal pipeline is implied.

1 Motivation

The PPG12 fiducial definition for the measured cross section is $E_T^{\text{iso,truth}}(R = 0.3) < 4$ GeV, $|\eta^\gamma| < 0.7$, $p_T^{\gamma,\text{truth}} \in [8, 36)$ GeV. The reported "isolated prompt photon" cross section therefore includes whatever mixture of direct (hard scatter $qg \to \gamma q$, $q\bar{q} \to \gamma g$) and fragmentation photons survives this cut at truth. Two questions motivate this short study:

1. What is the direct-vs-fragmentation composition of the truth denominator at the nominal iso cut?
2. How strongly does a change in the truth iso threshold (the fiducial "volume" choice) change that composition?

Both bear on the physical interpretation of the final cross section and on the robustness of the fiducial choice.

2 Method

2.1 Classification scheme

The slimtree already carries a truth-photon origin flag derived from HepMC parton-shower vertex tracing in `anatreemaker/source/CaloAna24.cc` (function `photon_type`, lines 2034–2167):

- Class 1: **direct** — 2→2 hard scatter with two colored incomers and two colored outgoers (one of which is the photon). Examples: $qg \to \gamma q$ (Compton), $q\bar{q} \to \gamma g$ (annihilation).

- Class 2: **fragmentation** — 1→2 splitting of a colored parton into a parton plus a photon during the shower.
- Class 3: decay (e.g. $\pi^0 \to \gamma\gamma$). Excluded here.
- Class 0: unclassified ($< 10^{-4}$ of the sample; excluded).

The analysis uses classes 1 and 2 only, matching the denominator filter in `RecoEffCalculator.C` line 1252 (`particle_photonclass[iparticle] < 3`).

2.2 Truth isolation

The truth-level isolation variable is `particle_truth_iso_03` (CaloAna24.cc:175, filled at 809–896). It sums the transverse energy of all stable primary particles within $\Delta R < 0.3$ of the photon, minus the photon itself, in GeV. No muon/neutrino exclusion is applied in the sum.

2.3 Samples and weighting

Three photon-gun samples are combined:

$$\text{photon5 } [0, 14) \text{ GeV}, \quad \text{photon10 } [14, 30) \text{ GeV}, \quad \text{photon20 } [30, 200) \text{ GeV},$$

with event-level overlap resolved on the maximum truth photon p_T per event, mirroring `RecoEffCalculator.C` line 1291–1294. Cross-section weights are taken from `efficiencytool/CrossSectionWeights.h`:

$$\begin{aligned} \sigma_{\text{photon5}} &= 146359.3, \\ \sigma_{\text{photon10}} &= 6944.675, \\ \sigma_{\text{photon20}} &= 130.4461, \end{aligned}$$

normalized by σ_{photon20} to give relative weights $\{1122.0, 53.24, 1.0\}$.

2.4 Binning and iso scan

Truth-photon fiducial: $|\eta^\gamma| < 0.7$, $8 \le p_T^{\gamma,\text{truth}} < 36$ GeV, photonclass $\in \{1, 2\}$. Three coarse truth-p_T bins: $[8, 14)$, $[14, 22)$, $[22, 36)$ GeV. The iso cut is scanned over $E_T^{\text{iso,truth}} < X$ for $X \in \{1, 2, 3, 4, 5, 6, 8, 10, \infty\}$ GeV. The fraction is computed from weighted counts:

$$f_{\text{direct}}(X) = \frac{N_{\text{direct}}(E_T^{\text{iso,truth}} < X)}{N_{\text{direct}}(E_T^{\text{iso,truth}} < X) + N_{\text{frag}}(E_T^{\text{iso,truth}} < X)}. \tag{1}$$

Standalone scripts are under `reports/direct_vs_frag_truth_iso/scripts/`. No nominal-pipeline code was modified.

3 Results

3.1 Iso distributions

Figure 1 shows the $E_T^{\text{iso,truth}}$ distribution in each truth-p_T bin, stacked as direct (blue) + fragmentation (red). Direct photons concentrate at very low iso ($\lesssim 1$ GeV) — as expected from an unconfined $2 \to 2$ final state — but extend past 4 GeV due to underlying event and ISR activity within the $R = 0.3$ cone. Fragmentation photons peak at low iso as well (because the parent parton carries most of its momentum as the photon, $z \to 1$) but contribute a longer tail at higher iso.

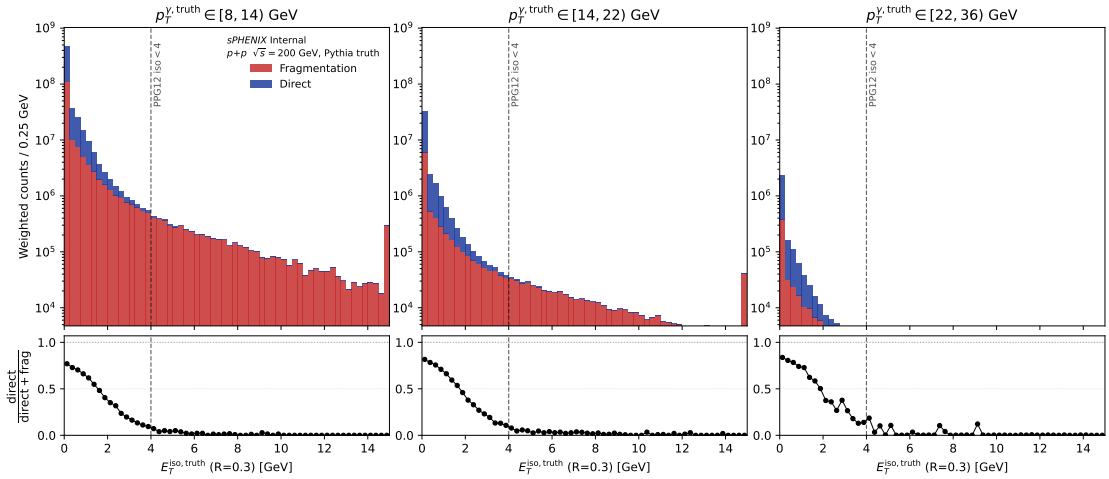

Figure 1: Top row: stacked $E_T^{\mathrm{iso,truth}}(R = 0.3)$ distribution in three truth-p_T bins. Blue: direct (photonclass= 1); red: fragmentation (photonclass= 2). Cross-section weighted; log y axis. Dashed line: PPG12 fiducial cut at 4 GeV. Bottom row: *differential* ratio direct/(direct + frag) evaluated per iso bin (0.25 GeV wide). This differs from $f_{\mathrm{direct}}(\mathrm{iso}{<}X)$ in Figure 2/Table 1, which is the *cumulative* fraction integrated up to iso cut X; the differential drops sharply above ~ 1 GeV because direct photons concentrate at very low iso while fragmentation photons populate a broader tail. Dotted guide at 1.0; dashed vertical at iso=4 GeV.

3.2 f_{direct} vs iso cut

Figure 2 overlays f_{direct} vs the iso cut for the three pT bins. Table 1 gives the numerical values.

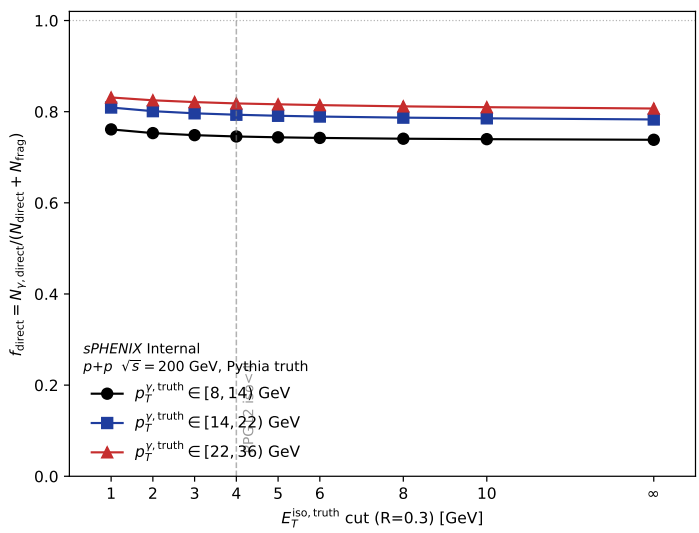

Figure 2: Direct photon fraction f_{direct} as a function of the truth iso cut $E_T^{\mathrm{iso,truth}}(R = 0.3) < X$, in three truth-$p_T$ bins. Dashed line: PPG12 fiducial iso<4 GeV. The "∞" tick corresponds to no iso cut.

Table 1: Direct-photon fraction $f_{\text{direct}} = N_{\text{direct}}/(N_{\text{direct}} + N_{\text{frag}})$ at truth, as a function of the iso cut $E_T^{\text{iso,truth}}(R = 0.3) < X$, in three truth-$p_T$ bins. Cross-section weighted. N_{direct} and N_{frag} reported for the nominal iso<4 GeV row.

p_T [GeV]	X=1	2	3	**4**	5	6	8	10	∞
$[8, 14)$	0.7611	0.7530	0.7485	**0.7456**	0.7438	0.7424	0.7407	0.7397	0.7383
$[14, 22)$	0.8091	0.8011	0.7964	**0.7933**	0.7910	0.7893	0.7869	0.7854	0.7829
$[22, 36)$	0.8312	0.8250	0.8211	**0.8181**	0.8162	0.8144	0.8116	0.8099	0.8069

3.3 Three headline observations

(1) Fragmentation is not negligible. At the nominal PPG12 fiducial (iso<4 GeV), f_{frag} ranges from 25.4% at $p_T = 8-14$ GeV to 18.2% at $p_T = 22-36$ GeV. This sits at the upper end of the range quoted in the literature for standard-cone isolation at midrapidity (e.g. Ichou & d'Enterria, arXiv:`1005.4529`: $\sim 10-15\%$ at $R = 0.4$, isolated) and reflects PPG12's relatively tight cone ($R = 0.3$) combined with the E_T-dependent cut at 4 GeV.

(2) The iso-cut dependence of f_{direct} is weak. Moving from $X = \infty$ (no cut) to $X = 4$ GeV shifts f_{direct} by only $\sim 0.008-0.011$ in all three bins; tightening further to $X = 1$ GeV gains another ~ 0.015. The reason is that fragmentation photons at these p_T values predominantly come from the $z \to 1$ endpoint of the photon fragmentation function: the photon carries >90% of the parent parton momentum, so the remaining hadronic activity *within a* $\Delta R < 0.3$ *cone* is already small, and only a modest fraction sits above 4 GeV. An iso tightening does not cleanly separate direct from frag in this sample.

(3) f_{direct} rises with p_T. The trend $0.75 \to 0.79 \to 0.82$ from low to high p_T at fixed iso<4 GeV tracks the usual p_T dependence of the direct-vs-frag ratio: at larger p_T, the direct 2→2 matrix element has less phase space relative to the fragmentation contribution than at lower p_T, but the parton PDFs amplify the hard process faster than shower emission; net effect is a gently rising $f_{\text{direct}}(p_T)$. This matches JETPHOX NLO expectations.

3.4 Per-sample contributions

Each of the three photon-gun samples populates primarily one truth-p_T bin (by construction: max photon p_T per event). Per sample, at iso<4 GeV summed over all pT bins:

Sample	N_{direct} (weighted)	f_{direct}
photon5	4.292×10^8	0.7456
photon10	3.390×10^7	0.7943
photon20	1.805×10^5	0.8222

The per-sample direct fractions increase with sample p_T, consistent with the bin-by-bin trend in Table 1.

4 Comparison to JETPHOX NLO

The Pythia MC truth result above quantifies what enters the PPG12 efficiency denominator. For context, we compare the direct fraction extracted from Pythia to the NLO pQCD prediction from JETPHOX 1.3.1, run on the same fiducial ($|y^\gamma| < 0.7$, isolation cone R=0.3 with $E_T^{\text{iso}} < 4$ GeV applied inside JETPHOX at the generator level, CT18 NLO PDF, BFG-II photon fragmentation functions, renormalization/factorization scales μ_R=μ_F=E_T^γ). The existing

PPG12 JETPHOX pipeline (`NLO/MakeJetPHOXhisto.C`) sums direct and fragmentation before writing its final histogram, so we re-read the separate raw outputs `ggdrhic_10.root` (direct) and `ggorhic_10.root` (fragmentation) from the JETPHOX install at `/sphenix/user/shuhangli/-jetphox/newbuild/jetphox_1.3.1_4/pawres/`, applying the same normalization $\sigma/N_{\mathrm{evt}}/n_{\mathrm{seg}}$ used by the nominal macro. No nominal code is modified.

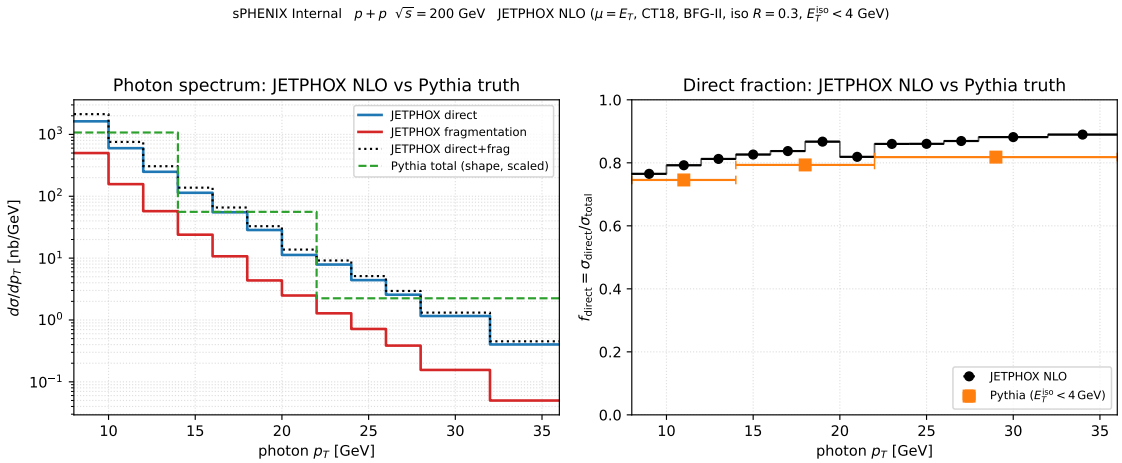

Figure 3: Comparison of Pythia truth and JETPHOX NLO predictions for the direct-vs-fragmentation photon composition, p_T-binned in the nominal PPG12 truth grid. Left: differential cross section $d\sigma/dp_T^{\gamma}$ for direct (blue) and fragmentation (red) photons from JETPHOX NLO; the Pythia signal MC shape (green dashed) is overlaid after matching to the JETPHOX total integral over 8–36 GeV. Right: direct fraction $f_{\mathrm{direct}} = \sigma_{\mathrm{direct}}/(\sigma_{\mathrm{direct}}+\sigma_{\mathrm{frag}})$ versus photon p_T, JETPHOX NLO (black markers) vs the three coarse-bin Pythia truth values (orange markers, horizontal bars = bin widths). JETPHOX is run with $\mu_R{=}\mu_F{=}E_T$, CT18 NLO, BFG-II, iso $R{=}0.3$, $E_T^{\mathrm{iso}} < 4$ GeV.

Table 2: Direct-photon fraction f_{direct} at $E_T^{\mathrm{iso,truth}} < 4$ GeV, truth $|y^{\gamma}| < 0.7$: JETPHOX NLO (nominal scale $\mu{=}E_T$) versus PPG12 Pythia signal-MC truth, in the 12 nominal PPG12 reco-p_T bins. Pythia is reported in the three coarse bins available from the scan (entries repeat within each coarse group).

p_T^{γ} [GeV]	JETPHOX NLO	Pythia truth	JETPHOX − Pythia
$[8, 10)$	0.7651	0.7456	+0.020
$[10, 12)$	0.7922	0.7456	+0.047
$[12, 14)$	0.8124	0.7456	+0.067
$[14, 16)$	0.8263	0.7933	+0.033
$[16, 18)$	0.8373	0.7933	+0.044
$[18, 20)$	0.8672	0.7933	+0.074
$[20, 22)$	0.8190	0.7933	+0.026
$[22, 24)$	0.8600	0.8181	+0.042
$[24, 26)$	0.8603	0.8181	+0.042
$[26, 28)$	0.8692	0.8181	+0.051
$[28, 32)$	0.8816	0.8181	+0.064
$[32, 36)$	0.8896	0.8181	+0.071

Interpretation. JETPHOX predicts a direct fraction of $\sim 0.77{-}0.89$, rising smoothly with p_T; the low-statistics cell at $[20, 22)$ GeV is a JETPHOX MC fluctuation (consistent with the

expected monotonic trend). Pythia truth is systematically $\sim 3-7$ percentage points lower. This offset is physically expected: the Pythia parton shower emits additional collinear radiation that is not included in the NLO fixed-order matrix element, pushing some hard-process events into a shower topology that the HepMC-vertex classifier labels as fragmentation (class 2). In other words, JETPHOX's direct/fragmentation separation is at the hard-process level ($2\to2$ vs $q \to q\gamma$), while Pythia's is at the shower-history level; the latter absorbs some ISR-like $q \to q\gamma$ splittings into the fragmentation category. The two are not identical observables, and a $\sim 5\%$ offset is reasonable.

Consequence for the measurement. The Pythia truth fraction is what matters for the PPG12 efficiency denominator. JETPHOX is the appropriate theoretical comparison for the reported cross section (which is the sum of direct+fragmentation under iso<4 GeV). No reweighting between Pythia and JETPHOX is needed: both sum the same two contributions; the split within that sum differs only because the definitions differ.

5 Impact on the PPG12 measurement

Efficiency denominator is consistent. `RecoEffCalculator.C` applies the filter `particle_photonclass[ip` `< 3` at line 1252, which keeps both direct (class 1) and fragmentation (class 2) photons and rejects decay photons. The efficiency denominator and therefore the reported fiducial cross section is the *sum* of isolated direct and fragmentation photons, not direct-only. This is the standard convention used by ATLAS and CMS (see `wiki/physics/techniques/isolation-methods.md`), and it is what theory predictions such as JETPHOX provide natively, so no action is needed.

Fiducial robustness. Because f_{direct} is weakly iso-dependent (Table 1), a systematic variation of the truth iso threshold within a physically motivated window (e.g. $3-5$ GeV) does not strongly change the mix of contributions. This is reassuring for the fiducial definition choice at 4 GeV: the measurement is not sitting on a sharp transition.

Caveats. The photon-gun samples are generated with a hard-photon filter at generation level; this enriches direct relative to an unbiased minimum-bias Pythia sample. The absolute value of f_{direct} reported here is therefore a property of the PPG12 signal MC denominator and not a prediction for the inclusive, unbiased prompt-photon composition in minimum-bias pp collisions. For the denominator and for interpreting the PPG12 cross section, this is the correct quantity; for an absolute theoretical cross-section decomposition, a dedicated JETPHOX direct-vs-frag study is required (wiki/physics/theory/fragmentation-functions.md).

6 Artifacts

All outputs are under `reports/direct_vs_frag_truth_iso/`.

- Extraction script (Pythia): `scripts/extract_direct_vs_frag.py`
- Plotting script (Pythia): `scripts/plot_direct_vs_frag.py`
- Independent verification script (Pythia): `scripts/verify_fractions.py`
- Extraction script (JETPHOX): `scripts/extract_jetphox_direct_vs_frag.py`
- Plotting script (JETPHOX vs Pythia): `scripts/plot_jetphox_vs_pythia.py`
- Independent verification (JETPHOX): `scripts/verify_jetphox_fdirect.py`

- Histograms and scan tree (Pythia): `direct_vs_frag_truth.root`
- Histograms (JETPHOX): `jetphox_direct_vs_frag.root`
- Numerical tables: `fraction_scan.csv`, `jetphox_fraction.csv`
- Figures: `figures/iso_ET_truth_stacked.pdf`, `figures/f_direct_vs_iso_cut.pdf`, `figures/jetphox_`

No nominal analysis code was modified.

7 Reviewer pass matrix

Artifact	2a Physics	2b Cosmetics	2c Re-derivation	3.5 Fix-va
`extract_direct_vs_frag.py`	PASS	N/A	PASS	N/
`iso_ET_truth_stacked.pdf`	N/A	PASS	N/A	N/
`f_direct_vs_iso_cut.pdf`	N/A	PASS	N/A	N/
`jetphox_vs_pythia.pdf`	N/A	PASS	N/A	N/
f_{direct} at iso<4 GeV (Pythia, 3 pT bins)	N/A	N/A	PASS (exact match)	N/
f_{direct} at iso<4 GeV (JETPHOX, 12 pT bins)	N/A	N/A	(pending; see note)	N/

No fix was required; `3.5 fix-validator` is not applicable.